



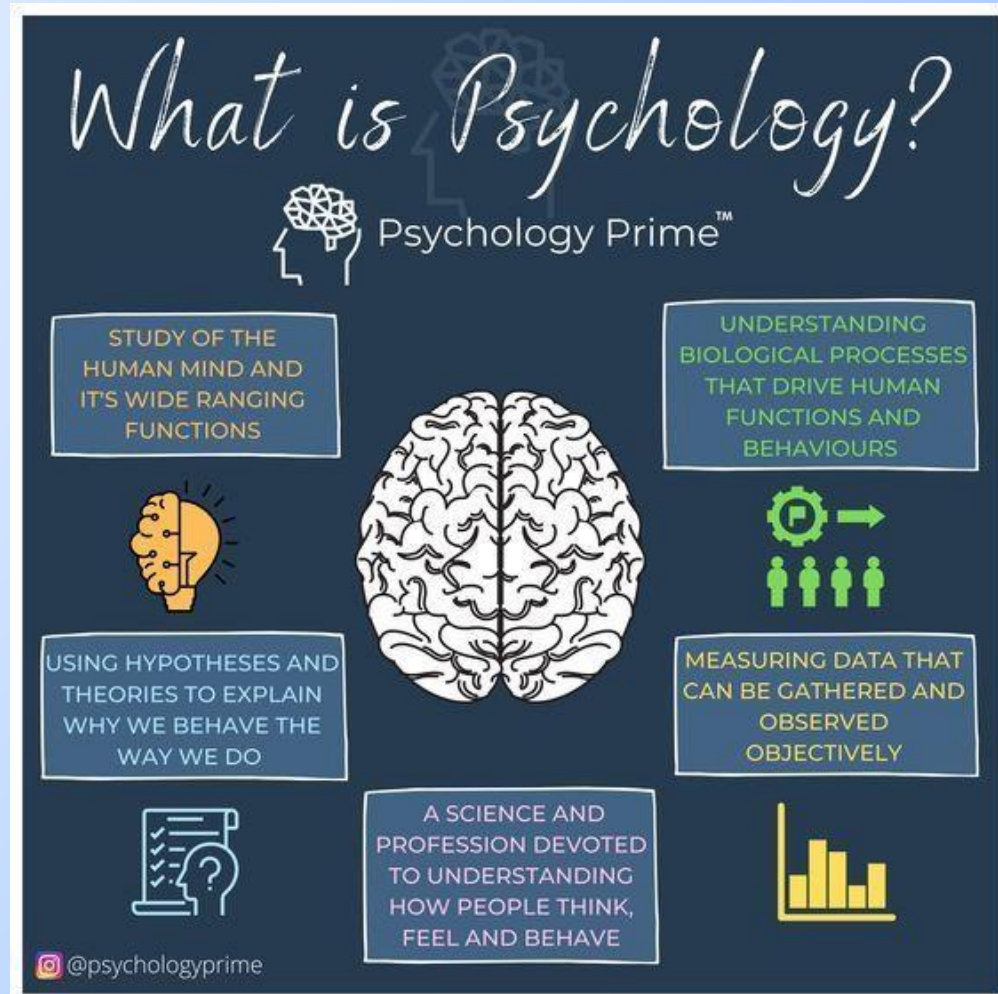
SCIENTIFIC FOUNDATIONS OF HEALTH

UNIT-1

Health and psychology

Psychology is the scientific study of the mind and behavior

Health and Well-Being describes psychological well-being as “the combination of feeling good and functioning effectively.”



What is Health Psychology

- Psychology is the scientific study of the mind and behavior.
- Health psychology is the study of psychological and behavioral processes in health, illness, and healthcare.
- It is concerned with understanding how psychological, behavioral, and cultural factors contribute to physical health and illness.
- Psychological study focuses on:
 - ✓ *health promotion and maintenance*
 - ✓ *the psychological aspects of the prevention and treatment of illness.*
 - ✓ *The analysis of health psychologists and attempt to improve the health care system and the formulation of health policy.*



Anxious



Naughty



Stubborn



Impatient



Sensitive



Intelligent



Nice



Nervous



Emotional



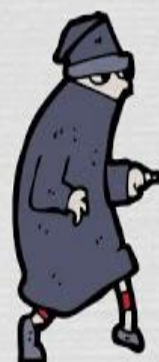
Bad-tempered



Mean



Distracted



Dishonest



Rude



Discreet



Crazy



Untidy



Cheeky



Cheerful



Energetic



Optimistic



Pessimistic



Unpleasant



Talkative



Calm



Passionate



Proud

Methods To Improve Good Psychological Health

1. Make social connections: especially face-to-face a priority

- Phone calls and social networks have their place, but few things can beat the stress-busting, mood-boosting power of quality face-to-face time with other people.

2. Stay active

- Staying active is as good for the brain as well as for the body. Regular exercise or activity can have a major impact on your mental and emotional health, relieve stress, improve memory, and help you sleep better.

Methods To Improve Good Psychological Health

3. Talk to someone

- Talk to a friendly face. In-person social interaction with someone who cares is one of the most effective ways to calm your nervous system and relieve stress.

4. Appeal to your senses

- Does listening to an uplifting song make you feel calm? What about taking a walk in nature and enjoying the sights and sounds of the trees? Everyone responds to sensory input a little differently, so experiment to find what works best for you.

5. Take up a relaxation practice

- Yoga, mindfulness, meditation and deep breathing can help reduce overall levels of stress.

Methods To Improve Good Psychological Health

6. Make leisure and contemplation a priority

Leisure time is a necessity for emotional and mental health. Take some time to relax, contemplate, and pay attention to the positive things as you go about your day — even the small things. Write them down if you can, because they can be easy to forget. Then you can reflect on them later if your mood is in need of a boost.

7. Eat a brain-healthy diet to support strong mental health

Foods that can support your mood include fatty fish rich in omega-3s, nuts (walnuts, almonds, cashews and peanuts), avocados, beans, leafy greens (spinach, kale and Brussels sprouts), and fresh fruit such as blueberries.

Methods To Improve Good Psychological Health

8. Don't skimp on sleep

It matters more than you think. One way to get sleep better is to take a break from the stimulation of screens — TV, phones, tablets or computers — in the two hours before bedtime.

9. Find purpose and meaning

- ✓ This is different for everyone. You might try one of the following:
 - Engage in work that makes you feel useful
- ✓ Invest in relationships and spend quality time with people who matter to you
- ✓ Volunteer, which can help enrich your life and make you happier
- ✓ Care for others, which can be as rewarding and meaningful as it is challenging

Psychological Disorders

- Psychological Disorder a “harmful dysfunction” in which behavior is judged to be (text discussion):
 - ✓ Atypical
 - ❖ not enough in itself
 - ✓ Disturbing
 - ❖ varies with time and culture
 - ✓ Maladaptive
 - ❖ harmful
 - ✓ Unjustifiable
 - ❖ By what standard?

Psychological Disorders

□ Neurodevelopmental Disorders

- Intellectual Disability
- Global Developmental Delay
- Communication Disorders
- Autism Spectrum Disorder
- Attention-Deficit Hyperactivity Disorder (ADHD)

□ Bipolar and Related Disorders

- Mania
- Depressive Episode

□ Anxiety Disorders

- Generalized Anxiety Disorder (GAD)
- Agoraphobia
- Social Anxiety Disorder
- Specific Phobias
- Panic Disorder
- Separation Anxiety Disorder

□ Stress-Related Disorders

- Acute Stress Disorder
- Adjustment Disorders

□ Dissociative Disorder

- Dissociative Amnesia
- Dissociative Identity Disorder



Psychological Disorders

□ Somatic Symptom Disorders

- Illness Anxiety Disorder
- Conversion Disorder

□ Eating Disorders

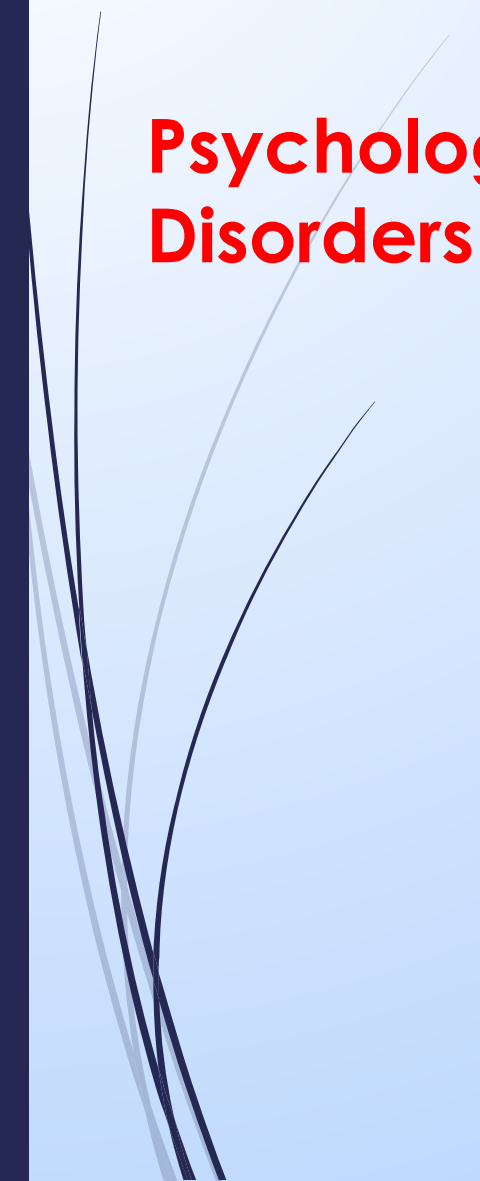
- Anorexia Nervosa
- Bulimia Nervosa
- Rumination Disorder
- Pica

□ Sleep Disorders

- Narcolepsy
- Insomnia Disorder
- Hypersomnolence
- Breathing-Related Sleep Disorders
- Parasomnias
- Restless Legs Syndrome



Psychological Disorders



□ Disruptive Disorders

- Intermittent Explosive Disorder
- Oppositional Defiant Disorder
- Depressive Disorders

□ Depressive Disorders

- Major depressive disorder:
- Dysthymia
- Substance/medication-induced depressive disorder

Mind : Defined as Function

Individual is not functioning adequately based on either his/her standards or according to significant others in the person's life.

□ Almost all the disorders we discuss have symptoms that everyone experiences. Diagnosis of disorder depends of intensity, length of time and how much it's impacting on the person.

□ Depression

□ Anxiety

□ Psychosis?

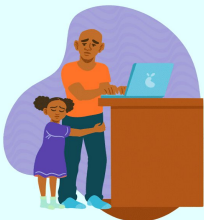
Medical Model

- concept that diseases have physical causes
- can be diagnosed, treated, and in most cases, cured
- assumes that these “mental” illnesses can be diagnosed on the basis of their symptoms and cured through therapy, which may include treatment in a psychiatric hospital

Anxiety Disorders

- distressing, persistent anxiety or maladaptive behaviors that reduce anxiety
- Generalized Anxiety Disorder
- person is tense, apprehensive, and in a state of autonomic nervous system arousal
- Persistence (out of control)
- Problem in identifying source

Children With Anxiety May...



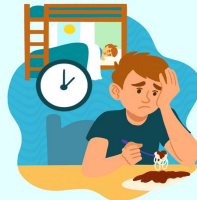
Appear more clingy than normal



Be restless and fidgety



Complain of stomachaches



Display changes in eating and sleeping habits



Express negative thoughts or worries



Get upset or angry more quickly



Have bouts of unexplained crying

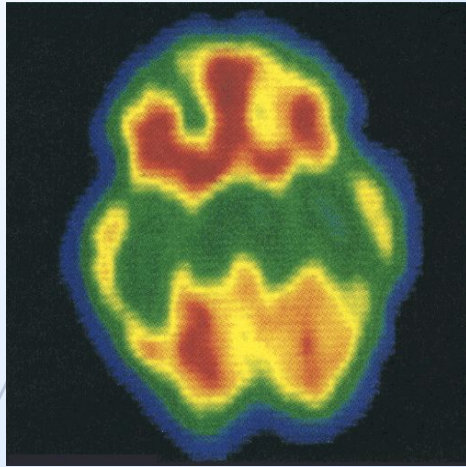


Struggle to concentrate

Anxiety Disorders / Panic Disorders

- marked by a minutes-long episode of intense dread in which a person experiences terror and accompanying chest pain, choking, or other frightening sensation
- Person comes to fear the panic attack itself and start to avoid any situations or places that might provoke an attack

Anxiety Disorders / Panic Disorders



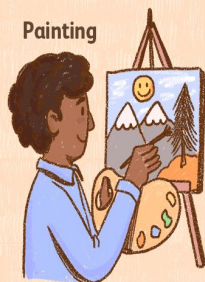
- PET Scan of brain of person with Obsessive/ Compulsive disorder
- High metabolic activity (red) in frontal lobe areas involved with directing attention (impulse control and executive function)
- Effectiveness of drug therapy



Strategies for Handling Anxiety



Exercising



Painting



Eating favorite foods



Meditation



Journaling



Watching funny online videos

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Tips for Coping with Anxiety

Focus on what you can change



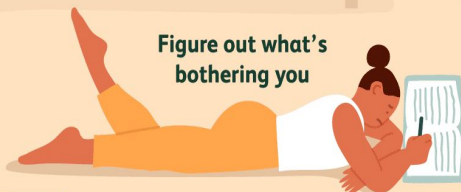
Stop and breathe



Focus on something less anxiety-provoking

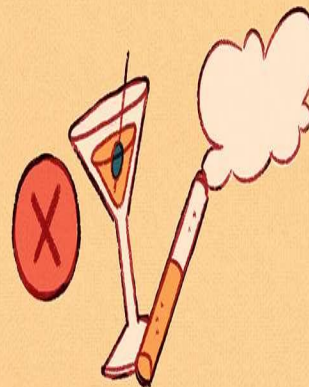


Figure out what's bothering you



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Lifestyle Changes for Lessening Symptoms of GAD



Limit alcohol, nicotine, and caffeine



Eat healthy foods



Exercise



Stay hydrated



Get plenty of sleep



Check all medications with your doctor

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Dissociative Disorders

- Dissociative Disorders
 - conscious awareness becomes separated (dissociated) from previous memories, thoughts, and feelings
 - Not uncommon when in a highly traumatic situation to feel “removed” from the situation. Problem is when this becomes more than a brief situation

Personality Disorders

- Personality Disorders (vs. mood disorder)
 - disorders characterized by inflexible and enduring behavior patterns that impair social functioning

Types of Personality Disorders

- Fearful, afraid of rejection, withdrawn
- Extreme eccentrics– “The Character”
- Narcissistic– Over exaggerates self importance
- Borderline– Unstable identity, emotions, relationships, etc.

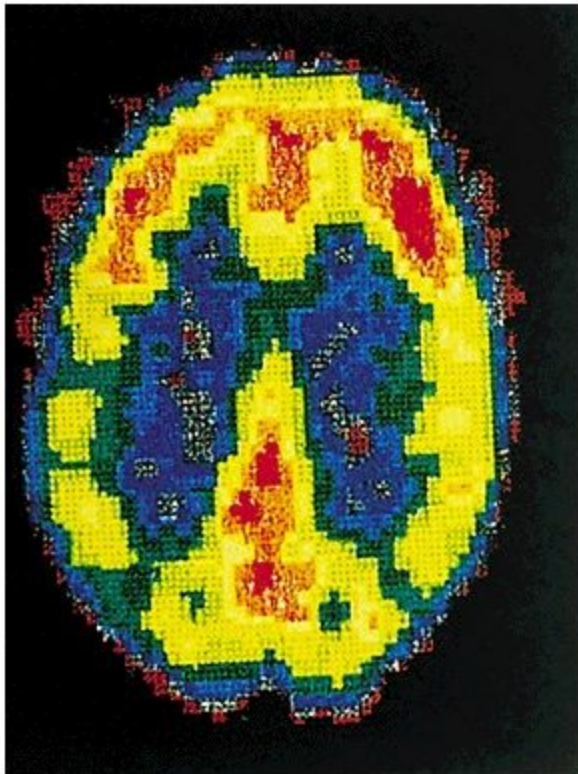
Personality Disorders

- Antisocial Personality Disorder
 - disorder in which the person (usually man) exhibits a lack of conscience for wrongdoing, even toward friends and family members
 - may be aggressive and ruthless or a clever con artist
 - Most criminals do not have this– they show concern for family and friends

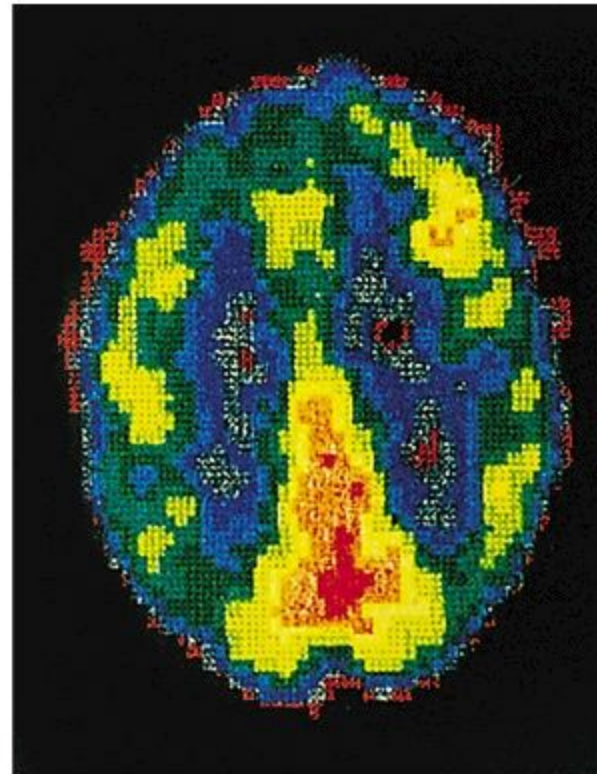
Personality Disorders

- PET scans illustrate reduced activation in a murderer's frontal cortex

Normal



Murderer



Mood Disorders

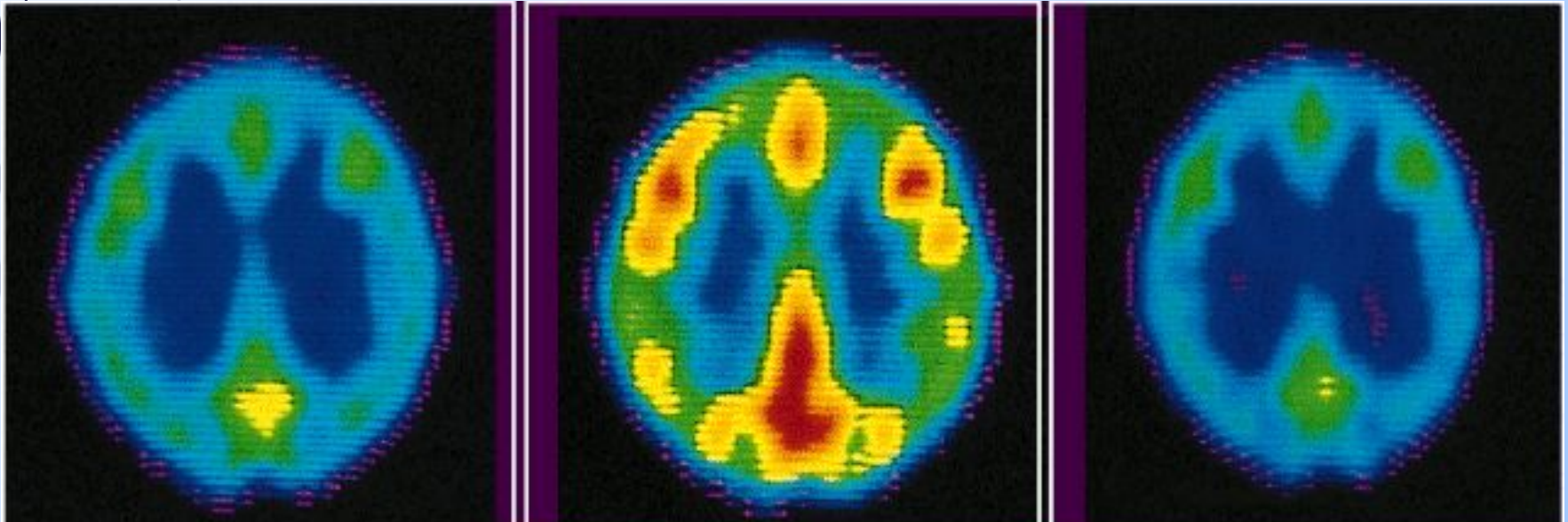
- Mood Disorders
 - characterized by emotional extremes
- Major Depressive Disorder
 - a mood disorder in which a person, for no apparent reason, experiences two or more weeks of depressed moods, feelings of worthlessness, and diminished interest or pleasure in most activities

Mood Disorders

- Manic Episode
 - a mood disorder marked by a hyperactive, wildly optimistic state
- Bipolar Disorder
 - a mood disorder in which the person alternates between the hopelessness and lethargy of depression and the overexcited state of mania
 - formerly called manic-depressive disorder

Mood Disorders-Bipolar

- PET scans show that brain energy consumption rises and falls with emotional switches

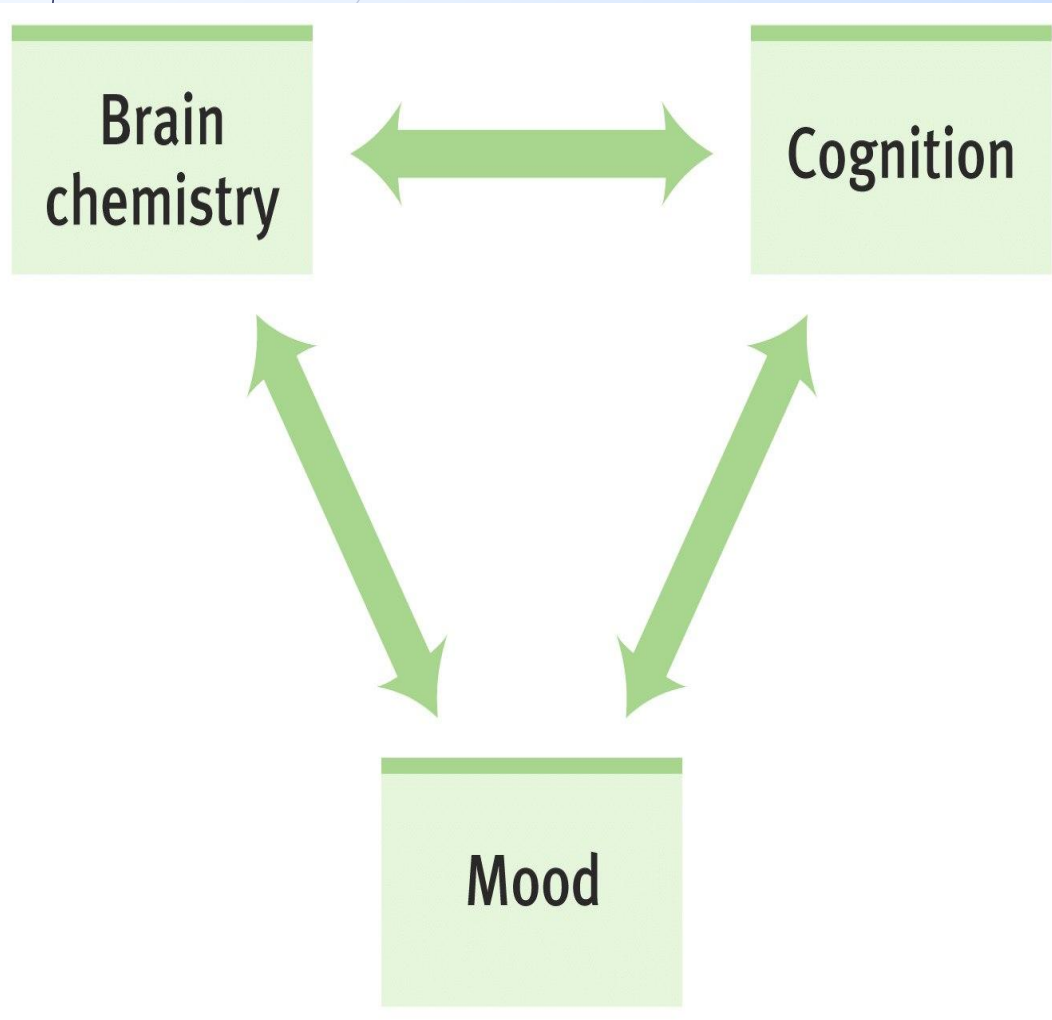


Depressed state

Manic state

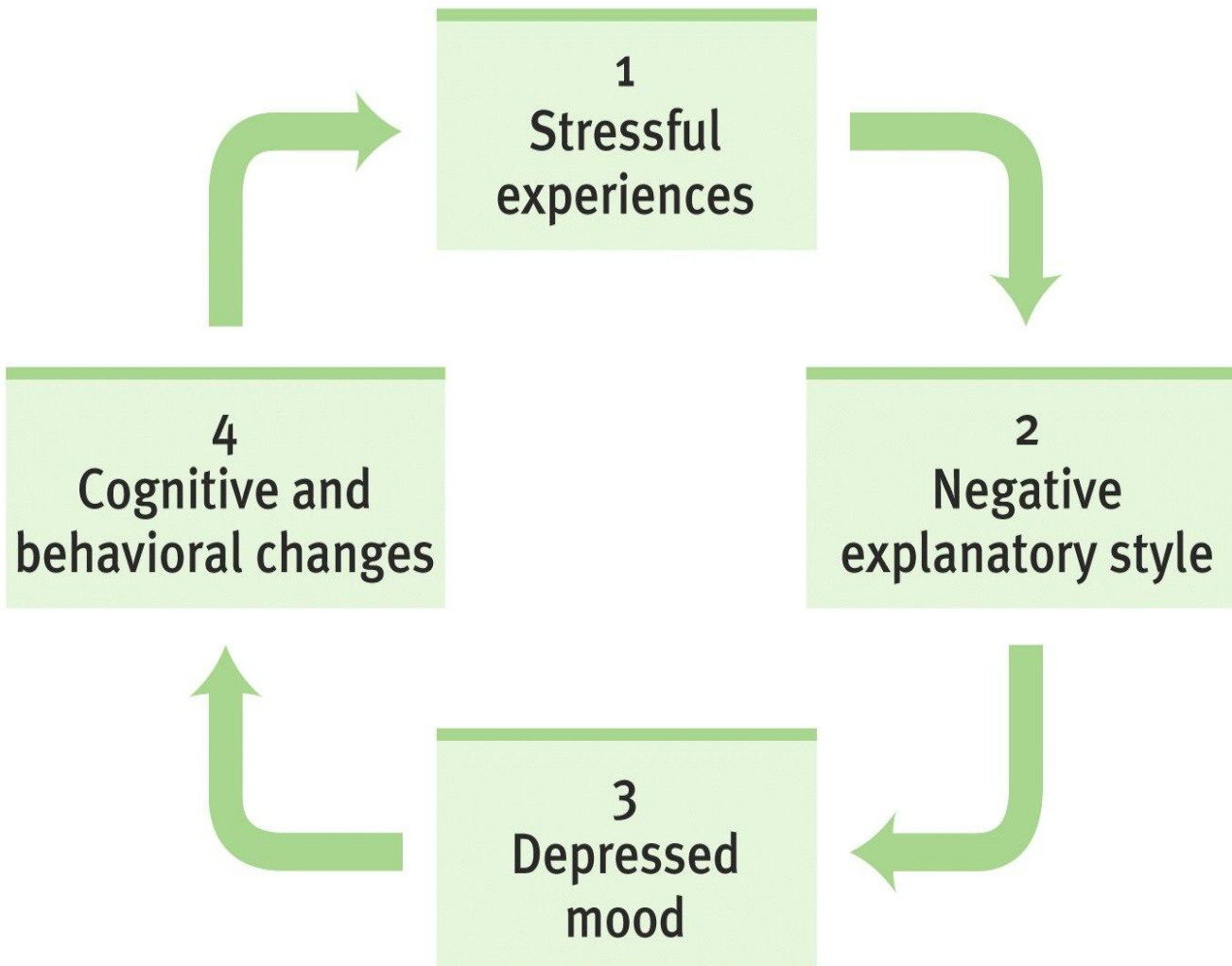
Depressed state

Mood Disorders-Depression



- Altering any one component of the chemistry-cognition-mood circuit can alter the others

Mood Disorders-Depression



- The vicious cycle of depression can be broken at any point

What is Stress?

- Stress can be defined as our mental, physical, emotional, and behavioral reactions to any perceived demands or threats.



The “Fight or Flight” Response

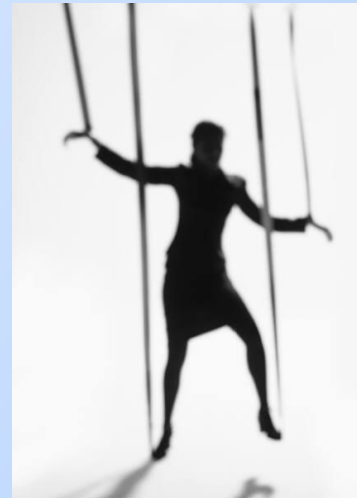


- When situations seem threatening to us, our bodies react quickly to supply protection by preparing to take action. This physiological reaction is known as the "fight or flight" response.
- The physiological response to a stressor is known as reactivity. Physiological responses can accumulate and result in long-term wear on the body.



What Makes Something Stressful?

- Situations that have strong demands
- Situations that are imminent
- Life transitions
- Timing (e.g., deviation from the “norm”)
- Ambiguity
- Desirability
- Controllability



Not All Stress is Bad...

□ **Distress** is a continuous experience of feeling overwhelmed, oppressed, and behind in our responsibilities. It is the all encompassing sense of being imposed upon by difficulties with no light at the end of the tunnel.

□ Examples of distress include financial difficulties, conflicts in relationships, excessive obligations, managing a chronic illness, or experiencing a trauma.

□ **Eustress** is the other form of stress that is positive and beneficial. We may feel challenged, but the sources of the stress are opportunities that are meaningful to us. Eustress helps provide us with energy and motivation to meet our responsibilities and achieve our goals.

□ Examples of eustress include graduating from college, getting married, receiving a promotion, or changing jobs.

Stress Response: Example

□ A good example of a stressful situation for many people is taking a test. If you find testing to be stressful, you might notice certain physical, behavioral, mental, and emotional responses.

- Physical Response?
- Behavioral Response?
- Mental Response?
- Emotional Response?



What is Stressful to You?

Work	Roommate	Legal matters
Classes	Childcare	Mental health
Studying	Finances	Law violation
Relationship with partner	Appearance	Spiritual/Religious issues
Relationship with family	Physical Health	Major/Career decisions
Relationship with friends	Not "fitting in"	Attitudes/thoughts
Trauma	Getting married	Buying a house
Change in residence	Change to a new school	Change in amount of recreation
Change in amount of social activities	Change in eating habits	Death of friend/family member

Why Do We "Stress Out"?

- For two major reasons:
 - We *perceive* a situation as dangerous, difficult, or painful.
 - We don't believe we have the *resources* to cope.



Stress Warning Signals

- What are your "red flags," or warning signs, that stress is creeping into your life? If we keep pushing ourselves, eventually something inside of us will send "red flags," or warning signs that stress is becoming a problem.



Suggestions for Reducing Stress

- 1. **Find a support system.** Find someone to talk to about your feelings and experiences.



Suggestions for Reducing Stress

- **2. Change your attitude.** Find other ways to think about stressful situations.
 - "Life is 10% what happens to us, and 90% how we react to it."



Suggestions for Reducing Stress



- **3. Be realistic.** Set practical goals for dealing with situations and solving problems.
 - Develop realistic expectations of yourself and others.

Suggestions for Reducing Stress

❑ **4. Get organized and take charge.** Being unorganized or engaging in poor planning often leads to frustration or crisis situations, which most always leads to feeling stressed.

❑ Plan your time, make a schedule, establish your priorities.



Suggestions for Reducing Stress

- **5. Take breaks, give yourself "me time."** Learn that taking time to yourself for rejuvenation and relaxation is just as important as giving time to other activities.
 - At minimum, take short breaks during your busy day.



Suggestions for Reducing Stress

- **6. Take good care of yourself.** Eat properly, get regular rest, keep a routine. Allow yourself to do something you enjoy each day.
 - Paradoxically, the time we need to take care of ourselves the most, when we are stressed, is the time we do it the least.



Suggestions for Reducing Stress

7. Learn to say "no."

Learn to pick and choose which things you will say "yes" to and which things you will not.

- Protect yourself by not allowing yourself to take on every request or opportunity that comes your way.



Suggestions for Reducing Stress



- ▣ **8. Get regular exercise.** Exercising regularly can help relieve some symptoms of depression and stress, and help us to maintain our health.



Suggestions for Reducing Stress

- ▣ 9. **Get a hobby, do something different.** For a balanced lifestyle, play is as important as work.



Suggestions for Reducing Stress

▣ **10. Slow down.** Know your limits and cut down on the number of things you try to do each day, particularly if you do not have enough time for them or for yourself.



- Be realistic about what you can accomplish effectively each day.
- Monitor your pace. Rushing through things can lead to mistakes or poor performance. Take the time you need to do a good job.
 - ◆ Poorly done tasks can lead to added stress.

Suggestions for Reducing Stress

▣ 11. Laugh, use humor.

Do something fun and enjoyable such as seeing a funny movie, laughing with friends, reading a humorous book, or going to a comedy show.



Suggestions for Reducing Stress

- **12. Learn to relax.** Develop a regular relaxation routine.
 - Try yoga, meditation, or some simple quiet time.



Relaxation Exercises

- Many different kinds, but 2 are:
 - **Deep Breathing**
 - **Visualization:** Visualization is a nice way of giving our minds and bodies a "mini vacation."





Stress Management: Next Steps

- Try to change the way you appraise a situation to make it less stressful
- Remember stress is normal but watch out for symptoms of stress
- Use coping skills/ways to reduce stress
- Practice relaxation techniques
- Contact TRiO advisor for counseling at 662-4365

Mindfulness for spiritual and Intellectual Health

Mindfulness exercises are **ways of paying attention to the present moment using techniques like meditation, breathing, and yoga**. Training helps people to become more aware of their thoughts, feelings, and body sensations so that instead of being overwhelmed by them, they're better able to manage them.

Why and how to change the brain

- Making life changes for health starts in the mind- getting out of auto pilot
 - Neuroplasticity- using your mind to change your brain to change your mind for the better!
 - Work on the mind will change your health Stress, pain, depression and anxiety

Mind Changes Brain

- What flows through the mind changes the brain physiologically
- Increased blood flow to active areas of brain
- Neurons that “Fire together wire together”
- Increasing excitability of active neurons
 - Building new synapses, thickening cortex
- Use it or lose it philosophy



Mindfulness

- 
- Attention is like a spotlight, illuminating what it rests upon
 - Directing attention skillfully is a fundamental way to shape the brain- over a lifetime
 - The brain shapes in increments, not dramatically, over time and practice

Changing health habits for good health

- ☐ Nutrition: Eat right
- ☐ Rest : Get optimum sleep
- ☐ Fitness: exercise regularly
- ☐ Manage social media time
- ☐ Medical check-ups
- ☐ Timely vaccination
- ☐ Control of substance abuse
- ☐ Stress Management
- ☐ Positive self talk
- ☐ Practicing gratitude
- ☐ Journaling
- ☐ Meaningful relationship with family and friends